Fiscal Year:	FY 2022	**Task Last Updated:**	FY 06/01/2022
PI Name:	Long, Marshall Ph.D.		
Project Title:	Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06520-8284	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	03/01/2017	**End Date:**	05/31/2022
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 05/31/2022 per NSSC information. (Ed., 1/31/22) NOTE: End date changed to 02/28/2022 per NSSC information. (Ed., 12/1/21) NOTE: End date changed to 8/31/2021 per D. Stocker/GRC and NSSC information (Ed., 4/20/2020) NOTE: Period of performance changed to 3/1/2017-2/28/2020 (original POP was 1/1/2017-12/31/2019) per D. Stocker/GRC (Ed., 1/17/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smooke, Mitchell Ph.D. (Yale University)		
Grant/Contract No.:	NNX17AD49A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Coflow Laminar Diffusion Flame" grant NNX11AP43A, with investigators Drs. Marshall Long and Mitchell Smooke. Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To understand better the factors that affect flame extinction and particulate formation, we propose to focus this study on flame phenomena at the extremes of the fuel dilution spectrum – namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames that have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform to gain a better understanding of the factors that affect diffusion flame lift-off distance and extinction, and it should also provide an excellent test-bed to determine the roles of inception, surface growth, and oxidation processes in soot formation models. The overall goals of the research are to improve our understanding of the physical and chemical processes controlling diffusion (i.e., non-premixed) flame structure and lifting phenomena (i.e., stabilization) and to provide for rigorous testing of numerical models, including thermal radiation, soot formation, and detailed chemical kinetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place.
Task Progress:	Coflow Laminar Diffusion (CLD) Flame had two main objectives as follows: A) Characterize the behavior of weak coflow diffusion flames methane and ethylene, as a function of velocity and fuel dilution. B) Characterize the sooting tendencies of methane and ethylene diffusion flames with long residence times, as a function of velocity and fuel dilution. Through the three rounds of testing of CLD Flame, both objectives were achieved. The first round of testing began in January 2018 and concentrated on measurements in dilute methane flames, as well as characterization and calibration of the imaging systems. The second round of experiments began in May 2018, and provided results on sooting methane and ethylene flames, as well as some data on weak, lifted ethylene flames. Because of issues related to the stability of the mass flow controllers, the quality of data on the extinction limits of the dilute methane and ethylene flames was not optimal. Therefore, a third round of experiments began in February of 2021. Improvements in the mass flow controllers allowed us to establish extinction limits for both fuels under a variety of flow conditions. The main product from CLD Flame is a comprehensive dataset of microgravity and normal gravity coflow methane and ethylene diffusion flames. These include weak, lifted flames near extinction, under a variety of fuel dilution levels and flow conditions. The highly diluted, lifted flames tend to be non-sooting and differences in liftoff height and flame stability between 1 g and 0 g are significant. This data will provide a stringent test for computational models seeking to predict flame extinction. At the other end of the fuel dilution spectrum, the dataset includes robust, sooting flames, with different flow conditions and dilution levels, which provide different flame heights and levels of soot loading. Once again, the contrast between 1 g and 0 g flames is remarkable and the lack of agreement between experimental and computational results points to deficiencies in current models of soot formation. The raw image data from CLD Flame has been processed to provide quantitative data on temperature in both the non-sooting and sooting flames using thin-filament and soot pyrometry, respectively. An absolute light calibration of the camera and optical train allows further quantification of CH* chemiluminescence in weak flames, as well as soot volume fraction in the case of sooting flames. The flow conditions are matched in the 1 g and 0 g datasets, and in both cases, detailed near-field velocity measurements have been made. These boundary conditions are critical as input to computational models wishing to compare experimental and predicted results. In addition to the raw data from 1 g and 0 g flames, processed data in the form of computer-readable data files, as well as final figures, have been uploaded to NASA.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Dobbins RR, Tinjero J, Squeo J, Zhao X, Hall RJ, Colket MB, Long MB, Smooke MD. "A combined experimental and computational study of soot formation in normal and microgravity conditions." Combust. Sci. Technol. 2022 Mar 29. https://doi.org/10.1080/00102202.2022.2041621 , Mar-2022